



Supervised Learning - Regression

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SUPERVISED LEARNING: REGRESSION

- Predicting continuous values from labeled examples.
- Examples: house prices, salaries, demand, costs.

From Data to Prediction





LEARNING OBJECTIVES

Explain regression

Distinguish linear vs polynomial regression

Describe train/test workflow

Interpret residuals

Evaluate with MAE, MSE, RMSE, R^2

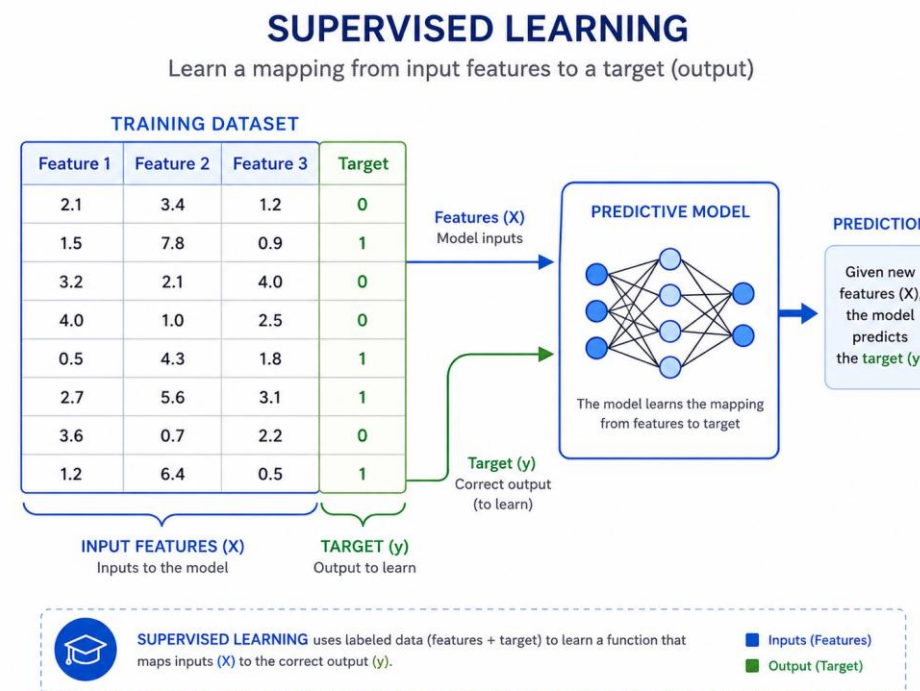


WHAT IS SUPERVISED LEARNING?

Model learns from examples with known answers.

- Input features X
- target value y

Goal: generalize to unseen data.





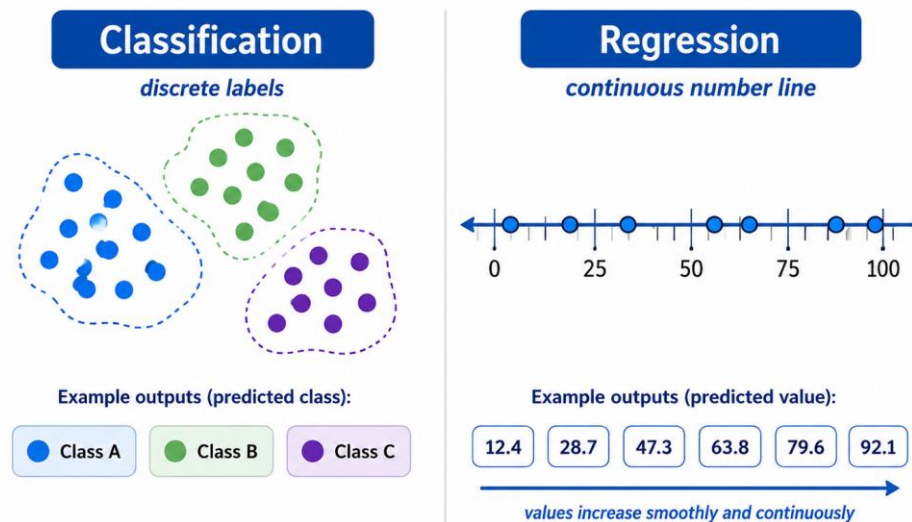
WHAT MAKES REGRESSION DIFFERENT?

Regression predicts numbers, not categories.

Classification: spam/not spam.

Regression: price, salary, temperature, time, demand.

Classification = discrete labels
Regression = continuous number line.

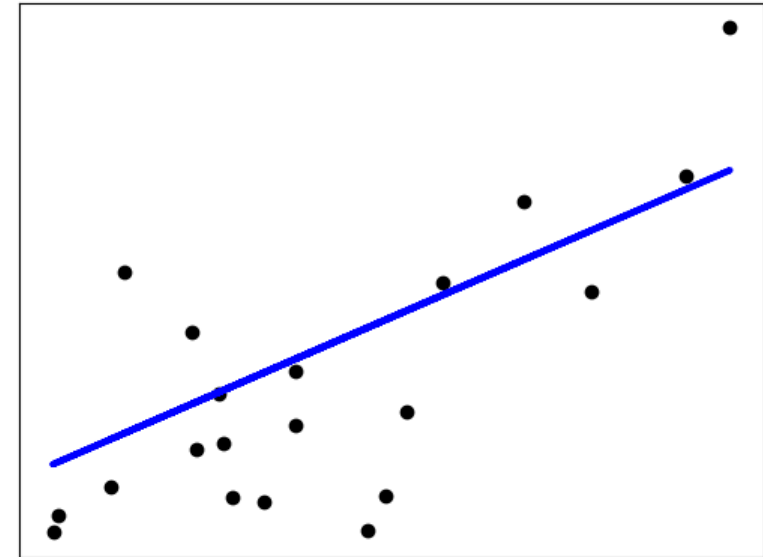




LINEAR REGRESSION INTUITION

Fit the best straight line through data.

The line summarizes a relationship between input and output.





THE REGRESSION EQUATION

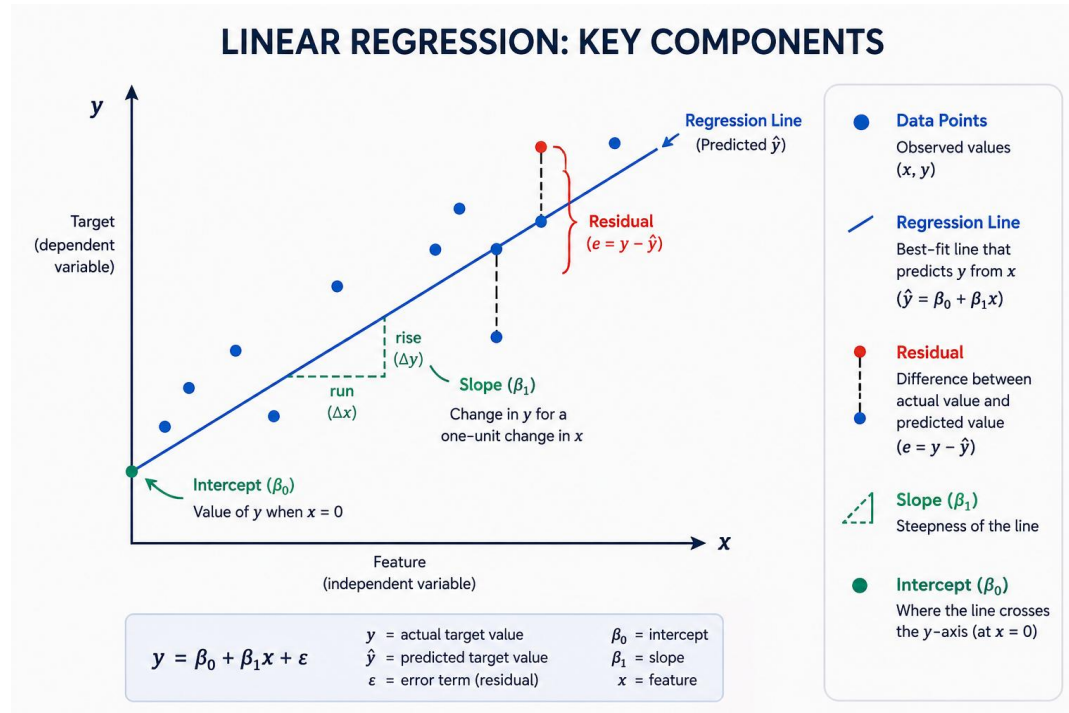
$$y = \beta_0 + \beta_1 x + \varepsilon$$

β_0 : intercept

β_1 : slope

ε : error/noise

Coefficients make linear regression interpretable.



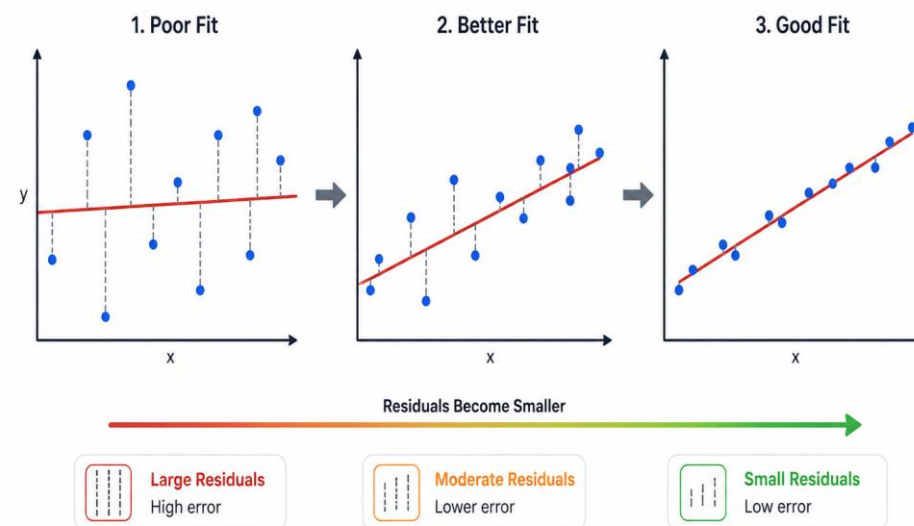


TRAINING THE MODEL

Training means finding coefficients that reduce prediction error.

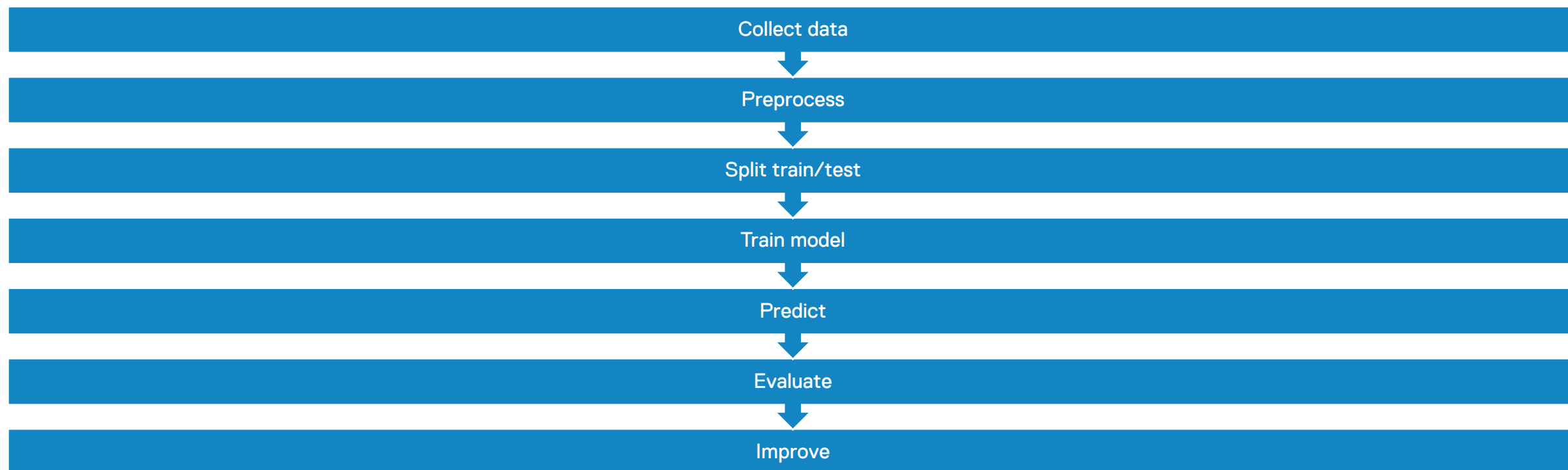
Ordinary Least Squares minimizes residual sum of squares.

Model Improving Over Time





REGRESSION WORKFLOW



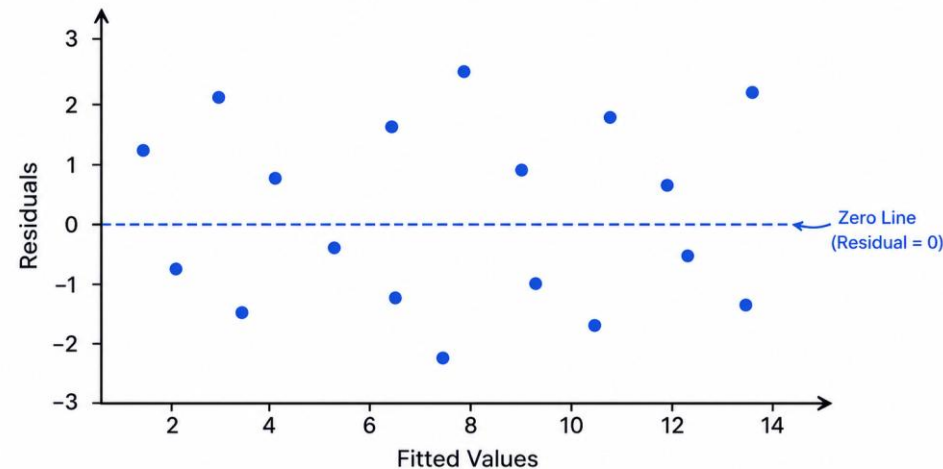


RESIDUALS: THE MODEL'S MISTAKES

- Residual = actual value – predicted value.
- Residual plots help detect bias, outliers, or non-linear patterns.

Residual Plot

Residuals vs. Fitted Values

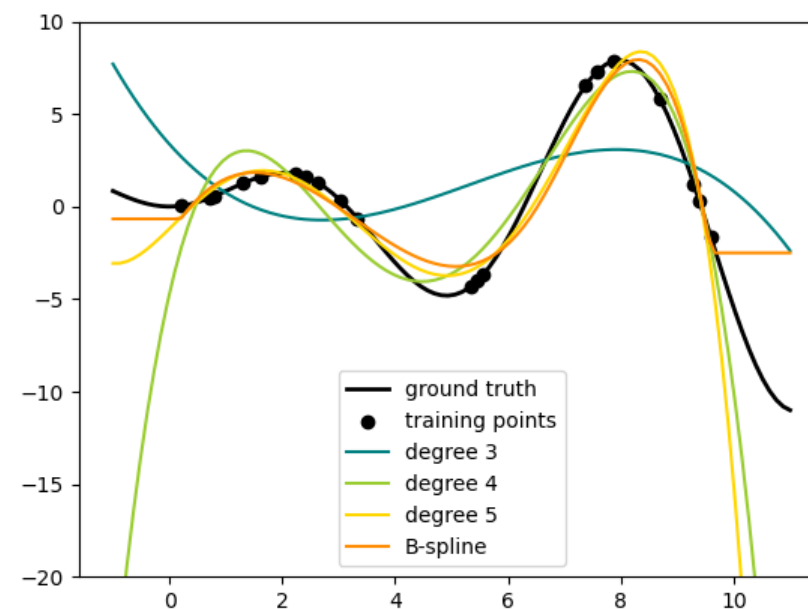


A good residual plot shows random scatter around the zero line with no clear pattern.



POLYNOMIAL REGRESSION

- Adds powers of features, such as x^2 , to model curved relationships.
- More flexible, but higher degree can overfit.



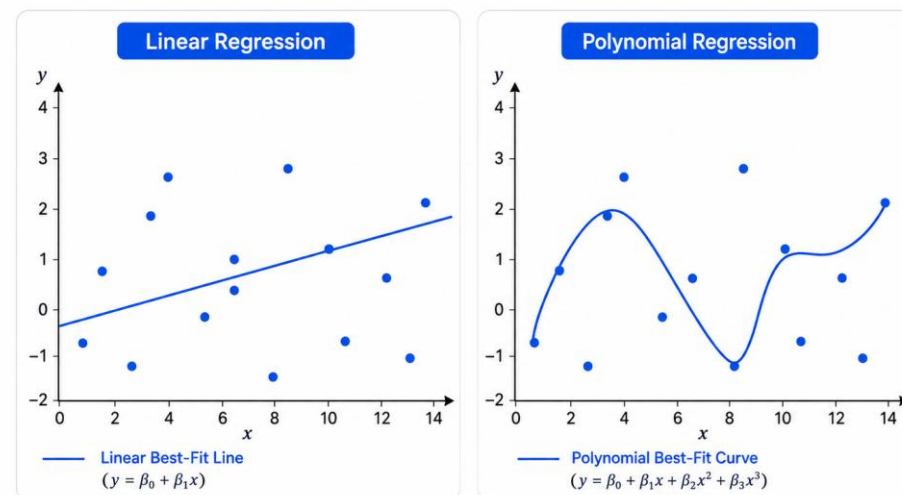


LINEAR VS POLYNOMIAL REGRESSION

- **Linear:** simple, interpretable, efficient.
- **Polynomial:** captures curves but can become unstable or too complex.

Linear vs. Polynomial Regression

Same Data, Different Models



Same data points (blue dots) are used in both models.
Linear regression fits a straight line. Polynomial regression fits a curve.



HOW DO WE EVALUATE REGRESSION?

- **MAE:** average absolute error.
 - **MSE:** squares large errors.
 - **RMSE:** error in target units.
 - **R²:** explained variance.
 - Use MAE for easy interpretation.
 - Use RMSE when large errors matter.
 - Use R² to communicate overall fit, but not alone.
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KEY TAKEAWAYS

- Regression predicts continuous values.
 - Linear regression is a strong baseline.
 - Polynomial regression adds flexibility.
 - Evaluation requires both metrics and plots.
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Technology Elevated